

Why a Thermocouple Reads Stably in Air but “Jumps” in Water: A Loop–Cutting View of Common–Mode and Ingress Failure Modes

A Technical Note

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Abstract

A thermocouple that reads steadily in air yet erratically once immersed is one of the most common bench complaints, and it is almost always first blamed on a waterproofing defect. We argue that the dominant cause is electrical rather than mechanical: conductive water closes a leakage–current loop that develops a *common–mode* voltage at the amplifier input, driving it out of its linear range. We present a unifying “loop–cutting” picture in which every standard remedy—full floating and galvanic isolation—is merely a different place to break the *same* loop, and we contrast this electrical family with the genuinely distinct case of water ingress forming a false junction. Five diagrams illustrate the mechanisms, and a single bench test is given that separates the two families.

1 The phenomenon

The thermoelectric (Seebeck) signal of a thermocouple is a differential voltage of only $\sim 40 \mu\text{V}/^\circ\text{C}$ for a type–K junction. Because it is so small, the measurement is acutely sensitive to any voltage that appears *common* to both leads. In still air the leads sit at a benign potential and the reading is stable. The instant the probe enters tap water—which is electrically conductive—a new current path appears, and the reading begins to wander. The reflex diagnosis, “the probe is not waterproof,” is usually wrong.

2 The leakage loop and the common–mode voltage

Figure 1 shows the mechanism. A mains–powered appliance sharing the bath (a heater, pump or circulator) leaks a small current into the water. The water carries that current to the thermocouple, up the leads, into the amplifier, and back to earth through the instrument’s ground, closing a loop. The leakage current i_ℓ flowing through the finite impedance to ground raises both leads together: a common–mode voltage

$$V_{cm} = i_\ell Z_{\text{gnd}}.$$

A differential front end rejects common–mode only while its inputs stay inside their specified range. Once V_{cm} pushes the inputs past the supply rails (or beyond the part’s stated common–mode window), the input protection conducts, common–mode rejection collapses, and the output is garbage. Because the leakage is mains–coupled, V_{cm} swings at 50/60 Hz; the sampler catches it at different phases, and aliasing between the conversion rate and the line frequency produces the characteristic intermittent “jumping” rather than a steady offset.

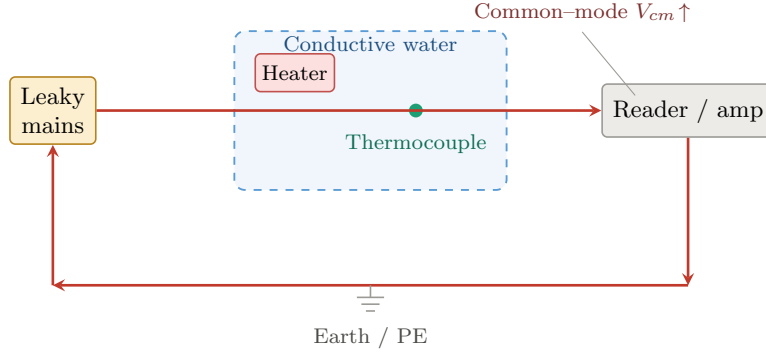


Figure 1: The leakage current (red) closes a loop and stacks a common-mode voltage V_{cm} across the amplifier input. Every remedy below is just a different place to cut this ring.

3 Remedy A: full floating

The most direct cut is to remove the amplifier’s connection to earth entirely: run the whole reader on a battery (Fig. 2). With nothing tying the instrument to earth, its reference is dragged along by the thermocouple so that the front end *rides* the water’s potential; the common-mode between the leads and the instrument’s own ground stays near zero, $V_{cm} \approx 0$. The weakness is fragility: connect a single earthed cable—a USB lead to an earthed PC, for example—and the cut is reconnected, so the cure holds only while the instrument is a true island.

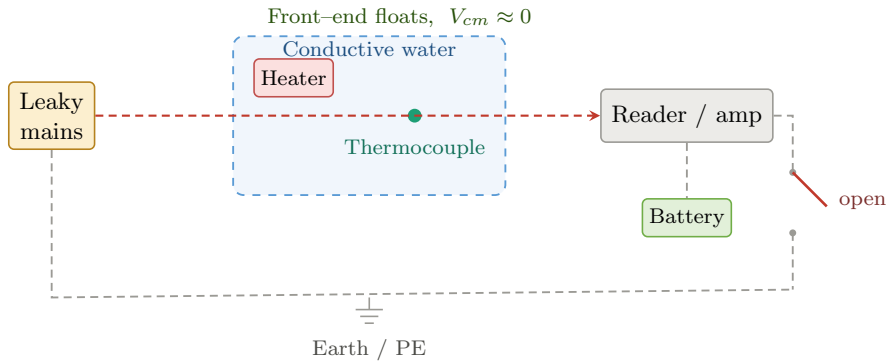


Figure 2: Full floating cuts the amplifier-to-earth segment. The loop cannot close, so $V_{cm} \approx 0$ —but one earthed cable re-closes it.

3.1 The “fake floating” trap: Y-capacitors

A switching DC adapter looks isolated but is not cut cleanly (Fig. 3). For EMI compliance it carries Y-capacitors to protective earth. These block DC but present a low impedance to mains-frequency AC, so the loop re-closes *weakly* through them. The common-mode is reduced but not zeroed—the familiar “much better, yet still jumps now and then.”

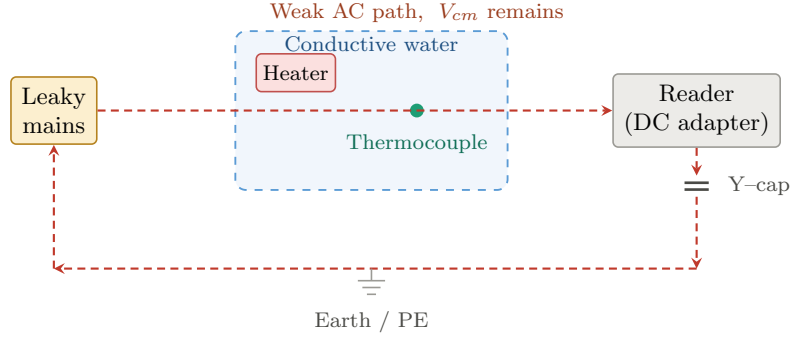


Figure 3: A Y-capacitor gives mains-frequency AC a weak path to earth, so the loop is only partly cut and residual V_{cm} survives.

4 Remedy B: galvanic isolation

The robust cut is made *inside* the amplifier (Fig. 4). An isolation barrier splits the part in two. The front end connects to the thermocouple and is free to float with the water; the back end stays earthed and may connect to a PC or any other earthed equipment. The conducted loop strikes the barrier and cannot cross. This is why isolation beats full floating: the back end can be tied to earth and to other gear without re-closing the loop.

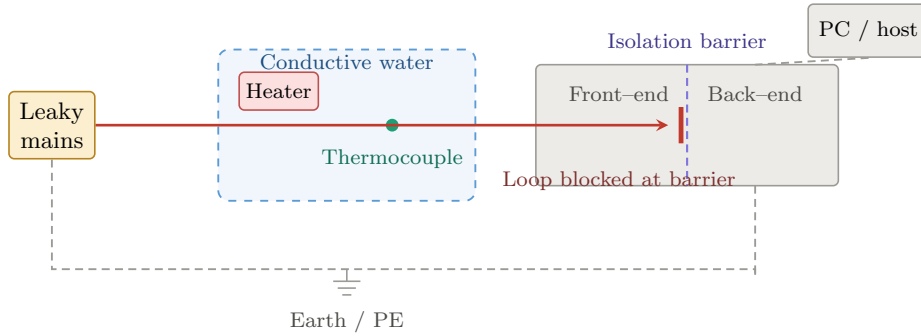


Figure 4: Galvanic isolation cuts the loop at a barrier inside the chip. The front end floats with the water; the earthed back end may connect freely.

5 A distinct failure mode: water ingress

The four cases above are one disease—a ground loop—cut in four places. The original suspicion, “not waterproof,” is a *different* disease with no relation to grounding (Fig. 5). A thermocouple reads wherever its junction is, normally at the sheath tip. If water penetrates a crack and bridges the two dissimilar wires partway along, that bridge becomes a new, wandering junction. The reading then reflects a moving, ill-defined point and jumps. This is “not waterproof” in the literal sense. Two further causes behave similarly by other mechanisms: unshielded leads picking up ambient EMI, and a loose or corroded connection.

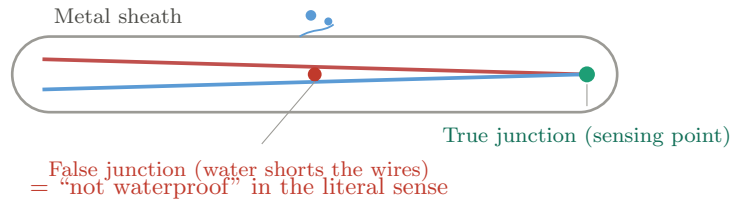


Figure 5: Ingress mode: water bridges the wires before the tip, creating a wandering false junction. Grounding fixes do nothing here.

6 Diagnosis and summary

The five figures collapse to two statements.

The first four are one disease—ground loop / common-mode. The red ring is the leakage current and V_{cm} is what it stacks up at the input. Full floating, the Y-capacitor case, and galvanic isolation differ only in *where* the ring is cut and *how cleanly*: full floating cuts at the amplifier-to-earth segment (clean but fragile); the Y-capacitor leaves an AC path (incompletely cut); isolation cuts at the in-chip barrier (robust, and it leaves the back end free to be earthed).

The fifth is a different disease—genuine water ingress, which forms a false junction and is untouched by any grounding measure.

A one-line bench test separates them: **wipe the probe dry and read it**. If it is stable immediately, the fault is electrical (one of the first four) and the seal is fine. If it recovers only after a long dry-out, water has gotten inside (the fifth). Two confirmatory checks: jumping that is synchronous with an appliance switching points to the loop or to transient coupling; jumping that changes when the leads are twisted or shortened points to differential pickup.